

$$T_1 \text{ 化简: } \overline{A}\overline{B} + C + \overline{A}\overline{C}D + B\overline{C}D$$

$$= (\overline{A} + B)\overline{C}D + \overline{A} + B + C$$

$$A + \overline{A}B = A + B$$

$$= \overline{A} + B + \overline{C}D + C$$

$$= \overline{A} + B + C + D$$

$$= \overline{A}\overline{B} + C + D$$

找反例

T_2

$$F = C + \overline{A}\overline{B}CD + A\overline{B}\overline{D} + \overline{B}\overline{C}\overline{D}$$

$$AD + BC = 0 \quad AD = 0 \quad BC = 0$$

0000	0101
0001	0100
0010	1000
0011	1000
1100	

$\overline{CD}$ $\overline{AB}$	00	01	11	10
00	1		1	1
01			1	1
11		X	1	1
10	1	X	1	1

使 $F=1$: $C=1$

$$A=1, B=0, D=0$$

得

$$C + \overline{B}\overline{D}$$

$$B=0, C=0, D=0$$

T₃

$$F = C\bar{A}\bar{B} + \bar{C}A\bar{B} + 0(\bar{A}B) + 1 \cdot (AB)$$

$$= AB + \bar{A}\bar{B}C + A\bar{B}\bar{C}$$

$$= A(B + \bar{B}\bar{C}) + \bar{A}\bar{B}\bar{C}$$

$$= A(B + \bar{C}) + \bar{A}\bar{B}\bar{C}$$

$$= AB + A\bar{C} + \bar{A}\bar{B}\bar{C}$$

T_p 假设 x, y 都是 2 位二进制已整教 $x, y \in [0, 3]$
 $z \in [0, 7]$

因此 z 可用 5 位二进制表示

$z_4 z_3 z_2 z_1 z_0$

输入

A₁A₀B₁B₀	A ₁ A ₀	B ₁ B ₀	z₄z₃z₂z₁z₀
0000	00	00	m ₁ 0 0 0 0 0
0001	00	01	m ₂ 0 0 0 0 0
0010	00	10	m ₃ 0 0 0 0 0
0011	00	11	m ₄ 0 0 0 0 0
0100	01	00	m ₅ 0 0 0 0 0
0101	01	01	m ₆ 0 0 0 0 1
0110	01	10	m ₇ 0 0 0 1 0
0111	01	11	m ₈ 0 0 0 1 1
1000	10	00	m ₉ 0 0 0 0 0
1001	10	01	m ₁₀ 0 0 1 0 0
1010	10	10	m ₁₁ 0 1 0 0 0
1011	10	11	m ₁₂ 0 1 1 0 0
1100	11	00	m ₁₃ 0 0 0 0 0
1101	11	01	m ₁₄ 0 1 0 0 1
1110	11	10	m ₁₅ 1 0 0 0 0
1111	11	11	m ₁₆ 1 1 0 1 1

$$Z_4 = \sum_m (15, 16)$$

$$Z_3 = \sum_m (11, 12, 14, 16)$$

$$Z_2 = \sum_m (10, 12)$$

$$Z_1 = \sum_m (7, 8, 15, 16)$$

$$Z_0 = \sum_m (6, 8, 14, 16)$$

$$Z_4 = A_1 A_0 B_1$$

$$Z_3 = A_1 \bar{A}_0 B_1 + A_1 A_0 B_2$$

$$Z_2 = A_1 \bar{A}_0 B_2$$

$$Z_1 = A_1 A_0 B_1 + \bar{A}_1 A_2 B_1$$

$$Z_0 = \bar{A}_1 A_0 B_2 + A_1 A_2 B_2$$